DRONACHARYA
College of Engineering

KHENTAWAS, FARRUKHNAGAR, GURGAON,

DEPARTMENT OF CSE (AI&ML)

FIRST SESSIONAL EXAMINATION (JANUARY-JUNE-2026)

(GURUGRAM UNIVERSITY)

Course Code: PCC-CSE-403	Course Title: Programming in JAVA	
Semester: IV	Maximum Marks: 70	Date: 12/03/2026
Branch :CSE(AI&ML)	Time: 03 Hours	Roll No.:

Note: : *Q.1 is compulsory. All questions are compulsory. Each question carries equal marks (14 marks). Assume data, if any are missing.*

1. ***Short Answer type questions based on REMEMBER/UNDERSTAND. (2 X 7=14 Marks***

a. Differentiate between class loading and class linking in java (CO-1)

b. State the difference between class variable and temporary variable (CO-1)

c. Identify the memory area where static methods are stored. (CO-2)

d. Give a brief explanation of method chaining in Java. (CO-1)

e. Describe the distinction between a class and an object in Java. (CO-1)

f. Discuss the method of passing arguments in Java. (CO-1)

g. Compare `super` and `this` keyword in Java. (CO-2)

SECTION-A

2 a) Write java program to count number of letters in a given string. **(CO-3)**

Input: -str1 = "Hello World" Output: - 10

b) Write java program to concatenate two strings into one string. **(CO-3)**

Input: - str1 = "cat", str2 = "dog", Output: - "cat dog"

OR

3. You are working on a project as a team lead with a team of 10 people. A developer uses variable name like I, j, k. What suggestion would you give him to name the variable names. What rules you prescribed him so that there is no difficulty for other developers to understand code. **(CO-1)**

4. You are given two files of java which are very bulky in size. How would you deploy these files? Write down the steps with example to deploy these files. **(CO-1)**

OR

5. A developer compile and execute his program. What happens when a java program is compiled and executed? What is the flow of java code? **(CO-1)**

SECTION-B

6. As a developer, you are working on a file named Mammal.java which requires some functionality, which is written in Animal.java. How would you access the functionality? Write clean code to describe your strategy with explanation. **(CO-2)**

OR

7. There is java file in which calculator method is defined class A. This method takes two integer argument and return their multiplication. As a developer, you required to multiply three numbers of float type. What strategy will you use in the class to multiply the numbers without altering the existing code and without changing method name. Write clean and full code and describe it. **(CO-2)**

8. How would you initialize instance variables with some values without using methods/functions? Write a clean and full code to describe the example. **(CO-1)**

OR

9. As a developer, you need to call constructor of parent class in child class. How would you do it. Write a clean code with small description. **(CO-1)**